

Dominic Bordelon

dominic.bordelon@gmail.com | <https://djbordelon.github.io/>

Summary

Adaptable research support specialist and information professional at a national research computing resource. International experience with a talent for engaging with varied domain experts. Unifies skills in data management and analysis, cultural competence, academic research, and technical communication with diverse stakeholders and clients.

Skills

Computer programming and communication: Python, SQL, R, data modeling, data validation, HTML, CSS, HTTP, Git, technical writing, Agentic AI use, Linux use

Analytics: cleaning/munging, data integration, basic statistical modeling, data visualization, web APIs

Interpersonal: information needs analysis, cultural competency, ethnographic interviewing

Languages: English, Spanish

Experience

Data Analyst – Curator, Pittsburgh Supercomputing Center – Pittsburgh, PA May 2024 – present

- Performs quality control checks and remedies on submissions to NIH-funded biomedical data repositories. Automates solutions using command-line tools. Communicates with data providers to acquire additional information and resolve issues.
- Translates information about repository holdings from local schemata to NIH standards for describing biomedical data for purposes of reporting and data discoverability.
- Operates processing pipelines on data submissions and checks results. Coordinates with pipeline developers, integration developers, the database administrator, and data providers to troubleshoot and resolve issues.
- Retrieves, manipulates, and visualizes repository metadata to inform stakeholders and guide decision-making.
- Develops software functionality for the Brain Image Library, enhancing the curation process and discoverability of the data.

Research Data Librarian, University of Pittsburgh – Pittsburgh, PA Nov 2019 – Apr 2024

- Consulted with faculty, post-doctoral fellows, and student researchers in a variety of disciplines about data management, computer programming, and research methods to help them solve problems in pursuit of their research (60 consultations in 2022)
- Developed and led introductory workshops for data science skills for the university community, mitigating computational-method gaps that exist in formal curricula (107 trainees in 2022)
- Created and hosted a weekly R and RStudio Drop-In Hour for the university community, enabling students to get free technical tutoring for assignments and projects (24 attendees in 2022)
- Delivered talks for faculty about data management, data sharing, reproducibility, and data privacy to better inform their research proposals to major funders and their responsible research practices (79 attendees in 2022)

- Collaborated with faculty in delivering course instruction to students, presenting guest lectures and supporting group projects with mentorship, to enhance learning with additional perspectives and expertise

Library Technology Specialist, European Southern Observatory–Garching, Germany Oct 2015–Oct 2019

- Developed software and databases (LAMP stack) for tracking publications that used observatory data (~4,200 articles for 2016-19) as a research tool for astronomers and an impact tool for data-informed decision-making by observatory administrators
- Implemented an intranet service (HTTP REST API) with internet landing pages for observatory departments to register digital object identifiers (DOIs), resulting in citable persistent identifiers for observatory publications and data sets
- Designed and built a variety of dashboards and tools for managing resources of the Office for Science, greatly facilitating logistics for students, fellows, and visiting scholars
- Fulfilled information requests in a specialized and international setting, connecting astronomers and engineers with needed information for research and development

Computer Services Librarian, E. Baton Rouge Parish Library – Baton Rouge, LA May 2014 – Sept 2015

- Conceived and implemented makerspace programming using Arduino DIY electronics kits for the general public, guiding participants in engagement with STEM.
- Transferred and automated existing record-keeping workflows to save many staff-hours for reporting & to provide a robust database solution for financial data (MS SQL Server)
- Solved IT support cases, including within the library management system, to correct errors and increase patron satisfaction

Reference Librarian , E. Baton Rouge Parish Library – Baton Rouge, LA	Jun 2013 – May 2014
Tutor (Part-time) , Acad. Ctr. for Student-Athletes, LSU – Baton Rouge, LA	Aug 2012–May 2015
Library Technician , E. Baton Rouge Parish Library – Baton Rouge, LA	Jan 2011 – May 2013
Teaching Assistant , Dept. of English, LSU – Baton Rouge, LA	Aug 2009 – May 2010
Spanish Teacher , Port Allen High School – Port Allen, LA	Aug 2008 – May 2009

Education

Master of Library and Information Science – Louisiana State University	2013
Bachelor of Arts, Spanish – Louisiana State University	2008
Bachelor of Arts, History – Louisiana State University	2008
Bachelor of Arts, English, concentration: Writing & Culture – Louisiana State University	2007
Certificate of Completion, Agricultural Data Analytics – ELVTR	2026
Certified CyberAmbassador – National Science Foundation	2025
Carpentries Certified Instructor – The Carpentries	2020

Selected Publications

Turner, Morgan L., Thomas C. Smits, Tiffany S. Liaw, et al. “HuBMAP Data Portal: A Resource for Multimodal Spatial and Single-Cell Data of Healthy Human Tissues.” Version 2. Preprint, arXiv, June 29, 2026. <https://doi.org/10.48550/ARXIV.2511.05708>.

- Bordelon, Dominic. (2026, January 23). *Wikipedia, AI, and Clearinghouses for Local and Regional Cultural Information*. Louisiana Folklore Society Annual Meeting, Lafayette, LA. <https://doi.org/10.5281/ZENODO.18626030>.
- Akrap, A., D. Bordelon, S. Chatterjee, et al. "Report on Reproducibility in Condensed Matter Physics." *Physical Review B* 113, no. 11 (2026): 119601. <https://doi.org/10.1103/27h6-yghn>.
- Phillips, G. A., Bordelon, D., & Kapral, T. T. (2023). [*"Push That Button and See What Happens": Addressing Technology Anxiety in Library Digital Scholarship Pedagogy*](#). In *Digital Humanities Workshops*. Routledge.
- Bordelon, D. (2022, October 6). *Documentation literacy as a metacognitive skill in computer programming*. Midwest Data Librarian Symposium, Notre Dame University.
- Bordelon, Dominic. (2022). The New College Try: Enrolling in First-Year Science Courses as Embedded Librarianship. <https://doi.org/10.17605/OSF.IO/NUEX5>
- Kerzendorf, W. E., Patat, F., Bordelon, D., van de Ven, G., & Pritchard, T. A. (2020). Distributed peer review enhanced with natural language processing and machine learning. *Nature Astronomy*. <https://doi.org/10.1038/s41550-020-1038-y>
- Bordelon, D., Grothkopf, U., & Meakins, S. (2018). First Light for DOIs at ESO. *EPJ Web of Conferences*, 186, 04003. <https://doi.org/10.1051/epjconf/201818604003>
- Patat, F., Boffin, H. M. J., Bordelon, D., Grothkopf, U., Meakins, S., Mieske, S., & Rejkuba, M. (2017). The ESO Survey of Non-Publishing Programmes. *The Messenger*, 170, 51–57. <https://doi.org/10.18727/0722-6691/5055>
- Leibundgut, B., Bordelon, D., Grothkopf, U., & Patat, F. (2017). Scientific Return from VLT instruments. *The Messenger*, 169, 11–15. <https://doi.org/10.18727/0722-6691/5032>